



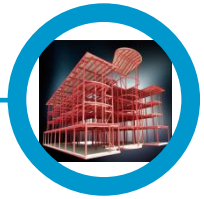
STEEL STRUCTURES/ SPACE STRUCTURES (CE-862) Fall Semester

Lecture 02-c



Welded Connections

Welded Connections



- **Welding Types & Methods**

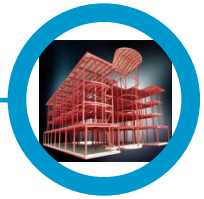
- AWS develops codes, recommended practices, and guides under strict ANSI procedures.

- D1.1 Structural Welding Code - Steel
  - One of the most consulted codes in the world.

- Most of what we do in structural steel design is derived from this specification!



Welded Connections



- **Welding Procedures**

- There are essentially four main types of welding methods used in structural applications:
  - Shielded Metal Arc Welding (SMAW)
  - Gas Metal Arc Welding (GMAW)
  - Flux Cored Arc Welding (FCAW)
  - Submerged Arc Welding (SAW)

- Each method has its advantages and disadvantages.
  - Let's look at each of these welding methods.

Welded Connections

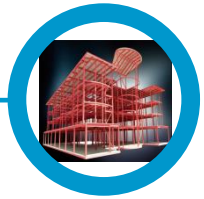
- **Welding Procedures**

  - Shielded metal arc welding (SMAW):
    - Also known as stick welding.
    - An electric arc is produced between the end of a coated metal electrode and the steel components to be welded.



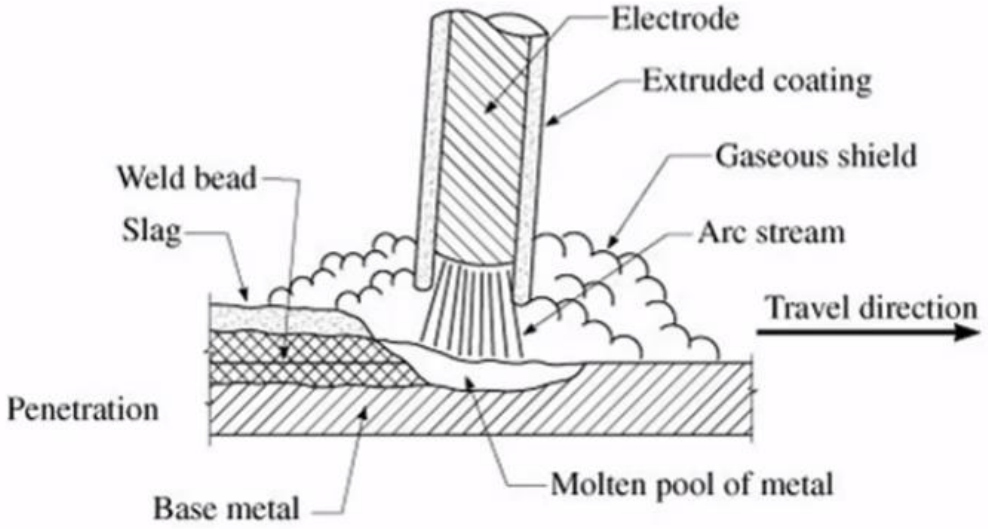


Welded Connections



- **Welding Procedures**

- SMAW Process:
  - The electrode is a filler metal covered with a coating.
  - The electrode's coating has two purposes:
    - It forms a gas shield to prevent impurities in the atmosphere from getting into the weld.
    - It contains a flux that purifies the molten metal.



Welded Connections

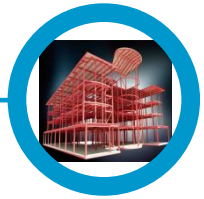
- **Welding Procedures**

- Gas metal arc welding (GMAW):
  - Also known as MIG welding.
  - Fast and economical.
- A continuous wire is fed into the welding gun:



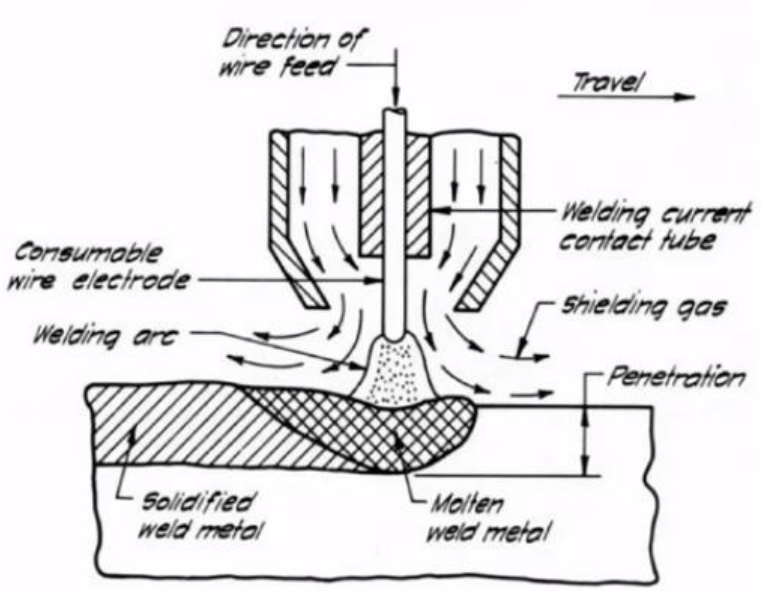


Welded Connections

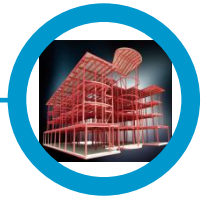


- **Welding Procedures**

- GMAW Process:
  - The wire melts and combines with the base metal to form the weld.
  - The molten metal is protected from the atmosphere by a gas shield which is fed through a conduit to the tip of the welding gun.
  - This process may be automated.



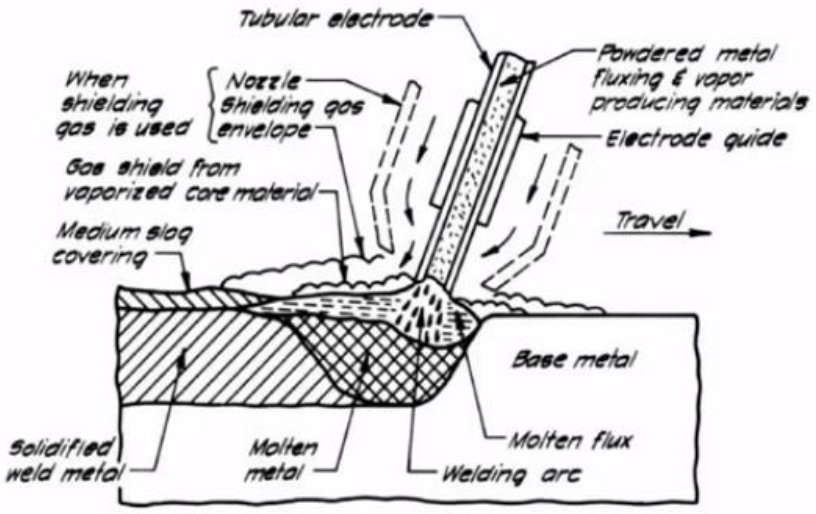
Welded Connections



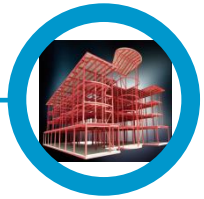
- **Welding Procedures**

- Flux cored arc welding (FCAW):
  - Similar to the GMAW process
  - A continuous wire is fed into the welding gun.

- The difference is that the filler wire has a center core which contains flux.
  - With this process it is possible to weld with or without a shielding gas.
  - This makes it useful for exposed conditions where a shielding gas may be affected by the wind.



Welded Connections

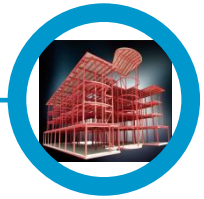


- **Welding Procedures**

- Submerged arc welding (SAW):
  - Only performed by automatic or semiautomatic methods
- Uses a continuously fed filler metal electrode.
  - Great for long, continuous welds.

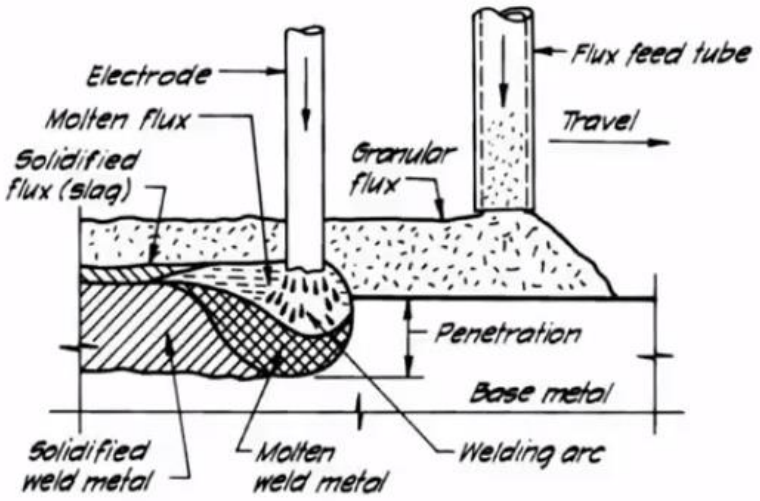


Welded Connections



- **Welding Procedures**

- SAW Process:
  - The weld pool is protected from the surrounding atmosphere by a blanket of granular flux fed at the welding gun.
    - Results in a deeper weld penetration than the other process
    - Only flat or horizontal positions may be used.



Welded Connections

- **Types of Welds**

- Tack welds:
  - A temporary weld used to hold parts in place while more extensive, final welds are made
  - More of a fabrication concern than a design one



Welded Connections

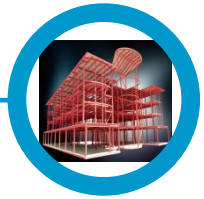
- **Types of Welds**

- Fillet welds:
  - The most common type of weld
    - Roughly 85% of all welded joints in structural applications are fillet welds
  - A bead of weld is placed along two adjacent members

- Fillet welds can be continuous welds or what are called stitch welds:
  - A series of shorter welds.



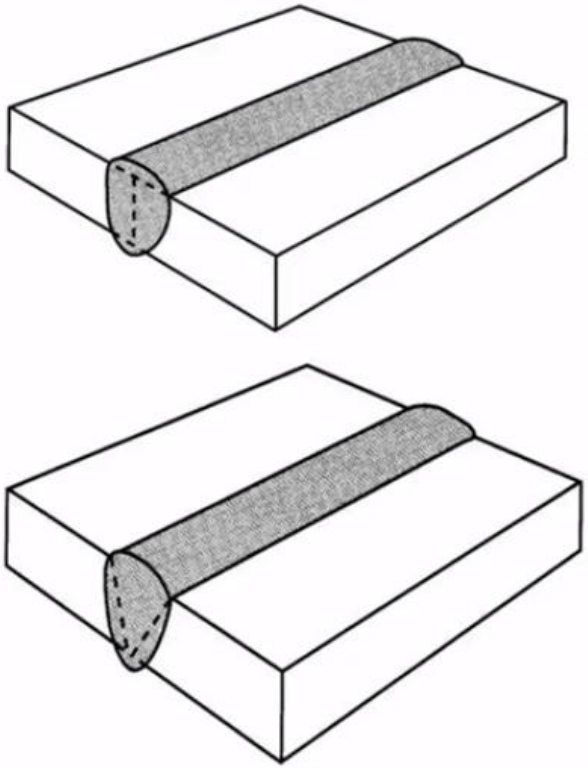
Welded Connections



- **Types of Welds**

- Groove welds:
  - A weld made in a groove between the work pieces.
  - Mostly used to splice or join two plates together.

  - Common types:
    - Complete Joint Penetration (CJP)
    - Partial Joint Penetration (PJP)



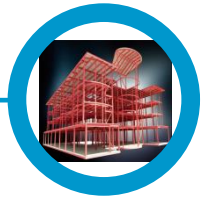
Welded Connections

- **Types of Welds**

- Plug/slot welds:
  - Not as common as fillet welds.
  - Can be used to:
    - Transmit shear in lap joints.
    - Join components of built-up members.
    - Prevent buckling of lapped parts.

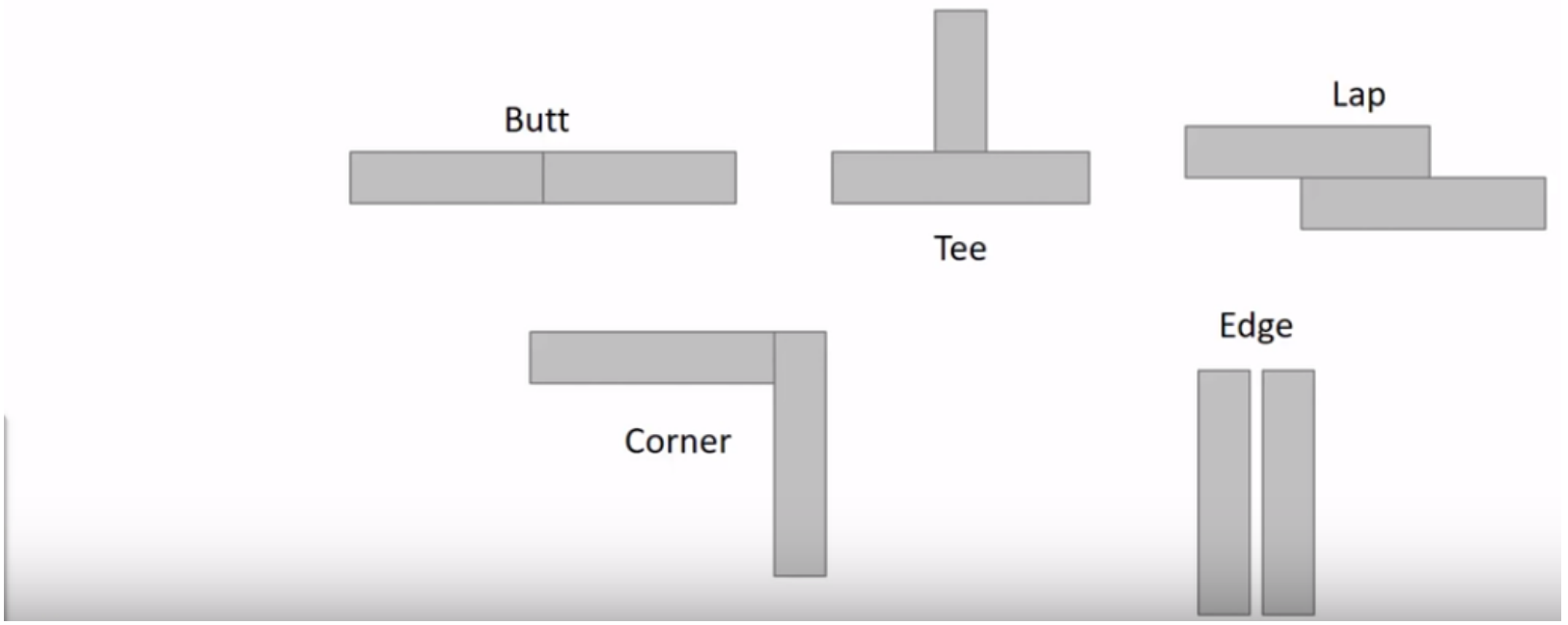


Welded Connections

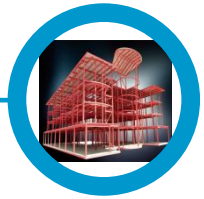


- **Types of Welds**

- Types of welded joints:
  - Lap, tee, and butt joints are the most common.

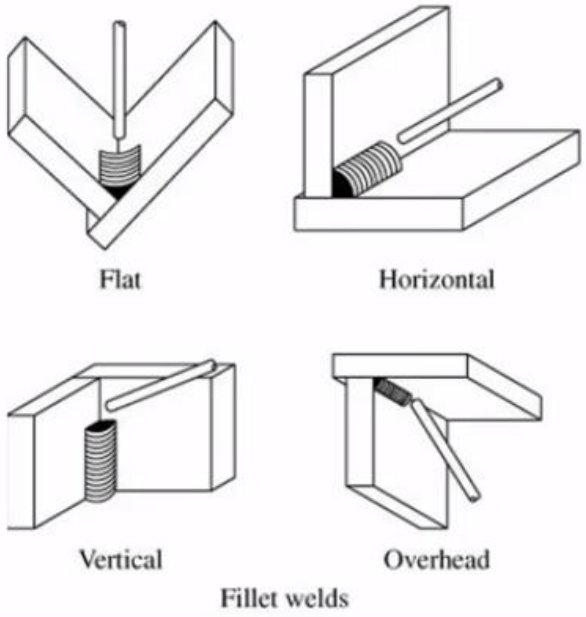


Welded Connections



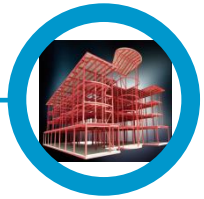
- **Types of Welds**

- Another important factor to consider is weld position.
  - The skill of the welder becomes quite critical for vertical and overhead welds.
  - When possible, we try to utilize only flat or horizontal positions.



Fillet welds

Welded Connections



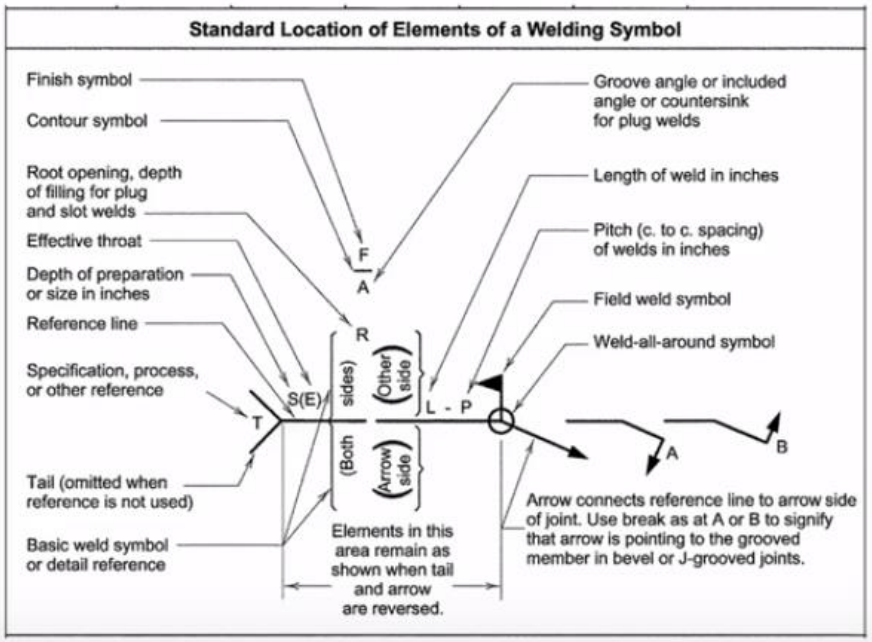
- **Types of Welds**

- Weld symbols & standard locations:

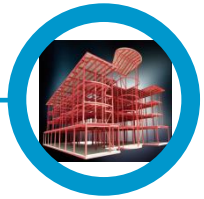
Table 8-2 (continued)
Prequalified Welded Joints

Basic Weld Symbols									
			Groove or Butt						
Back	Fillet	Plug or Slot	Square	V	Bevel	U	J	Flare V	Flare Bevel
									

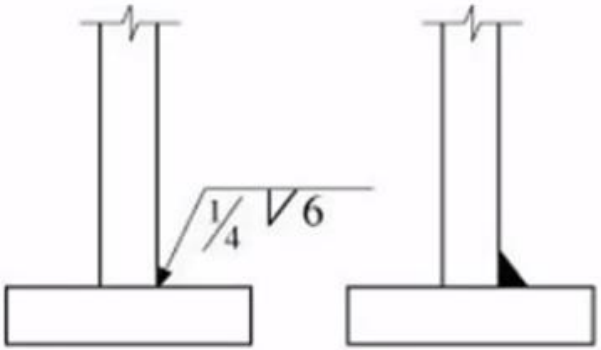
Supplementary Weld Symbols						
				Contour		
Backing	Spacer	Weld All Around	Field Weld	Flush	Convex	For other basic and supplementary weld symbols, see AWS A2.4
						



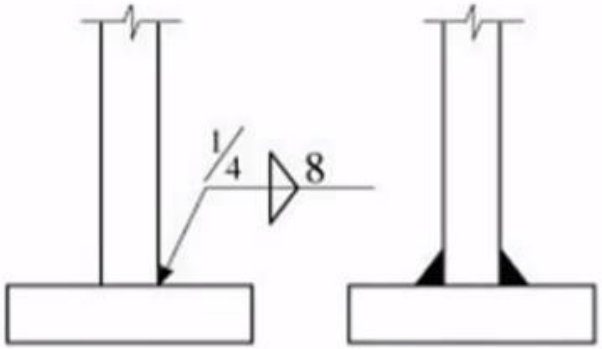
Welded Connections



- For this course, we'll only be covering fillet welds (most common).
  - Therefore, these are the only types of weld symbols that we'll use:

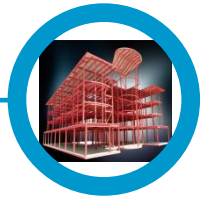


1/4-in. fillet weld, 6 in. long on near side (arrow side)



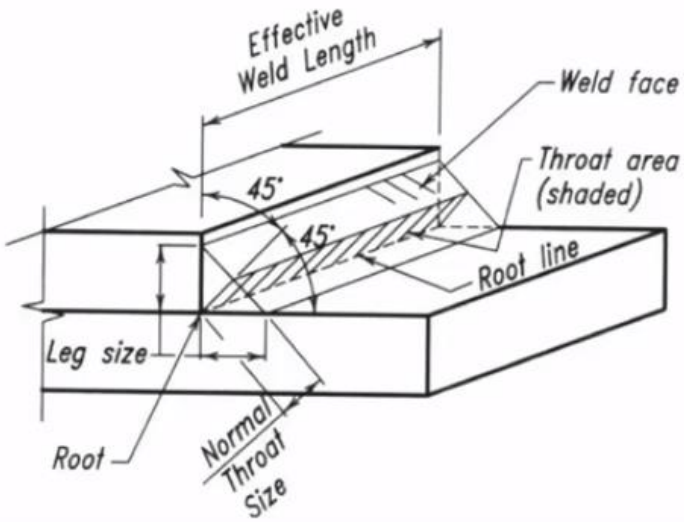
1/4-in. fillet weld, 8 in. long on both sides

Welded Connections



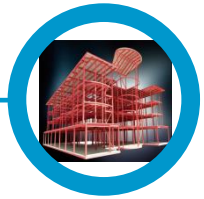
- **Fillet Weld Capacity**

- Weld capacity is traditionally defined based on the effective throat.
  - This is the section on a weld that is considered to be effective in carrying load.



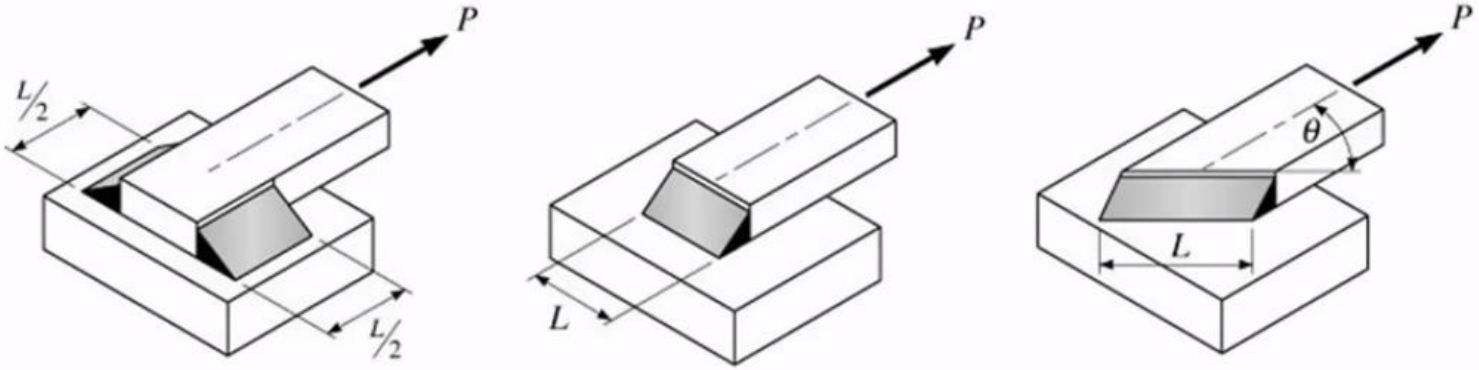
- One important concept:
  - Regardless of weld orientation, we always assume loads in welds to be transferred through shear!

Welded Connections

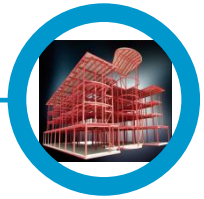


- **Fillet Weld Capacity**

- Again, we always consider the force on a fillet weld as a shear on the throat!
- However, how exactly does the orientation affect weld capacity?
  - Note that an angle of 0 degrees indicates pure shear on the weld!

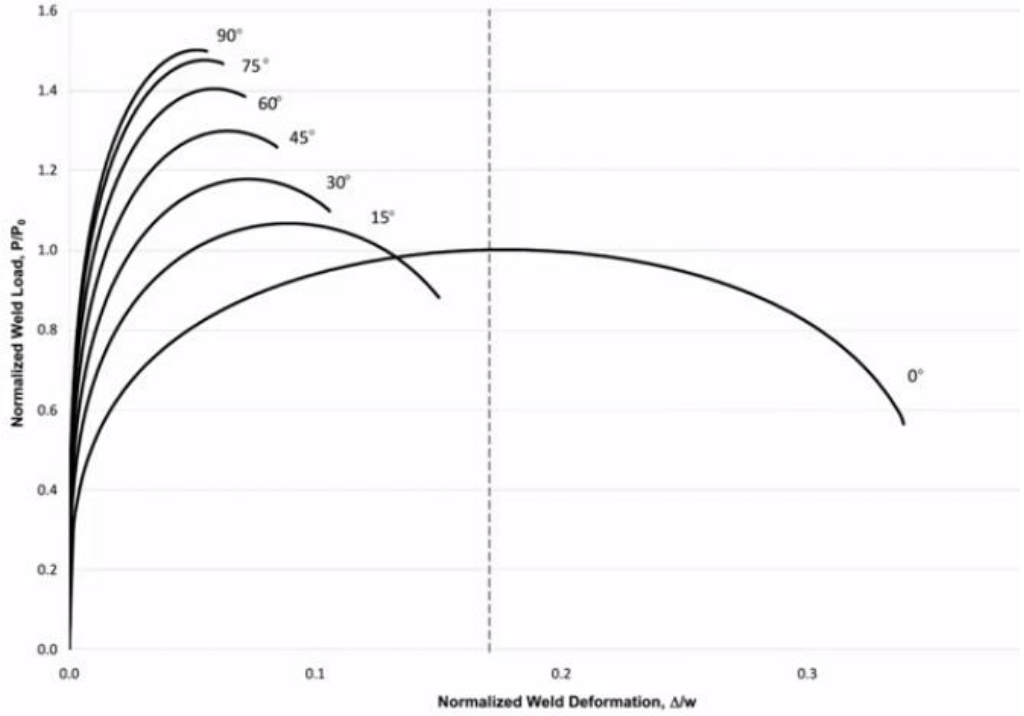


Welded Connections

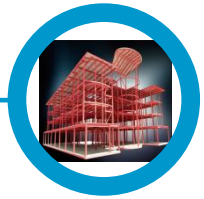


- **Fillet Weld Capacity**

- If we plot weld capacity as a function of orientation, we find:
  - It is conservative to assume welds are loaded in shear.
  - It is also advantageous!
    - More ductility when welds are loaded in shear!
  - Note: We NEVER use transverse welds alone!

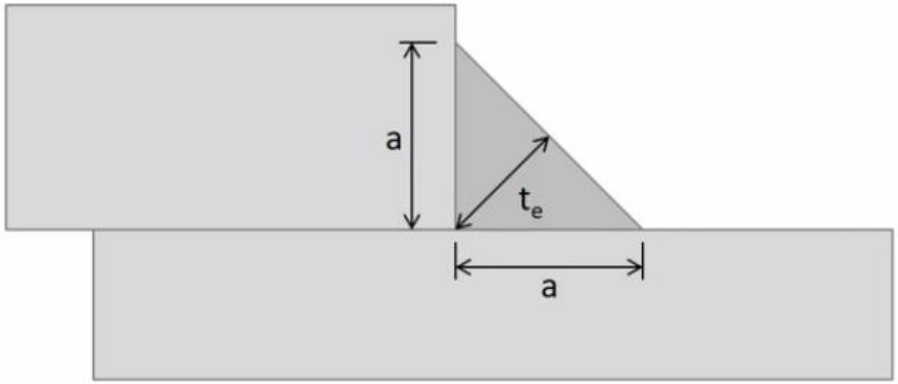


Welded Connections



- **Fillet Weld Capacity**

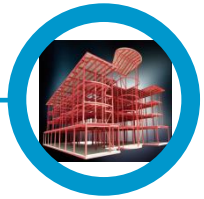
- Back to fillet weld capacity
  - If we define a fillet weld based on its leg size, “a”…



  - We can calculate the width of the effective throat, t_e, as shown:
    - 45° - 45° - 90° triangle

$$t_e = \frac{1}{\sqrt{2}} a \approx 0.707a$$

Welded Connections



- **Fillet Weld Capacity**

- Capacity of fillet welds is governed by the following equations:
  - Strength of the base metal.
  - Strength of the weld metal.

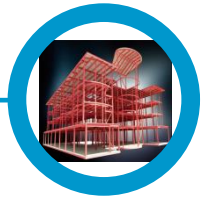
Pg-16.1-113

$$R_n = F_{nBM} A_{BM}$$ *Eqn. J2-2, Pg.*

$$R_n = F_{nw} A_{we}$$ *Eqn. J2-3, Pg.*

- F_{nBM} = nominal stress of the base metal (ksi)
- A_{BM} = area of the base metal (in^2)
- F_{nw} = nominal stress of the weld metal (ksi)
- A_{we} = effective area of the weld (in^2)

Welded Connections



- **Fillet Weld Capacity**

- If you review the Specifications…

Pg-16.1-113

For the weld metal

$$R_n = F_{nw}A_{we} \qquad \text{(J2-3)}$$

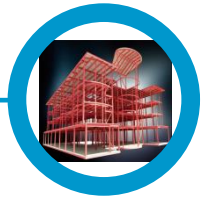
where

F_{nBM} = nominal *stress* of the base metal, ksi (MPa)
F_{nw} = nominal stress of the weld metal, ksi (MPa)
A_{BM} = cross-sectional area of the base metal, in.^{2} (mm^2)
A_{we} = effective area of the weld, in.^{2} (mm^2)

The values of ϕ, Ω, F_{nBM} and F_{nw} and limitations thereon are given in Table J2.5.

  - Let's look at this table.

Welded Connections



- **Fillet Weld Capacity**

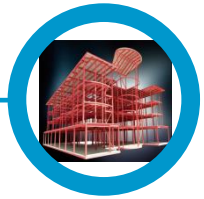
Pg-16.1-114

- From the Specifications…

TABLE J2.5 (continued)
Available Strength of Welded Joints, ksi (MPa)

Load Type and Direction Relative to Weld Axis	Pertinent Metal	ϕ and Ω	Nominal Stress (F_{nBM} or F_{nw}), ksi (MPa)	Effective Area (A_{BM} or A_{we}), in.^{2} (mm^2)	Required Filler Metal Strength Level[a][b]
FILLET WELDS INCLUDING FILLETS IN HOLES AND SLOTS AND SKEWED T-JOINTS					
Shear	Base	Governed by J4			Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_{EXX}$[d]	See J2.2a	
Tension or compression—Parallel to weld axis	Tension or compression in parts joined parallel to a weld is permitted to be neglected in design of welds joining the parts.				

Welded Connections



- **Fillet Weld Capacity**

- So, we now know two of the values:
  - Note: F_{EXX} = filler metal classification

$$\phi = 0.75 \qquad F_{nw} = 0.60F_{EXX}$$

- What about the area?

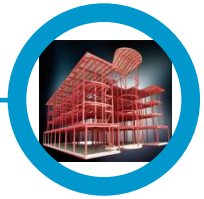
Sec J2.2a, Pg 16.1-110

$$\boxed{A_{we} = (0.707a)L_w}$$

2a. Effective Area

The effective area of a fillet weld shall be the effective length multiplied by the effective throat. The effective throat of a fillet weld shall be the shortest distance from the root to the face of the diagrammatic weld. An increase in effective throat is permitted if consistent penetration beyond the root of the diagrammatic weld is demonstrated by tests using the production process and procedure variables.

Welded Connections



- **Fillet Weld Capacity**

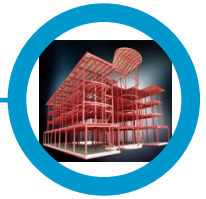
- Therefore, we can define the capacity of a fillet weld as follows:
  - Note: L_w = length of weld

$$\phi R_n = \phi F_{nw} A_{we}$$

$$\phi = 0.75 \qquad F_{nw} = 0.60 F_{EXX} \qquad A_{we} = 0.707 L_w a$$

$$\therefore \boxed{\phi R_n = \phi\left(0.60 F_{EXX}\right)\left(0.707 L_w a\right)}$$

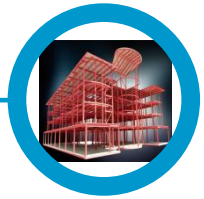
Welded Connections



- **Fillet Weld Capacity**

- This section is split into the following:
  - J4.1 Strength of Elements in Tension
    - Would indicate only a transverse fillet weld (bad idea!)
  - J4.2 Strength of Elements in Shear
    - What we're looking for
  - J4.3 Block Shear Strength
    - Should be familiar…
  - J4.4 Strength of Elements in Compression
  - J4.5 Strength of Elements in Flexure

Welded Connections



- **Fillet Weld Capacity**

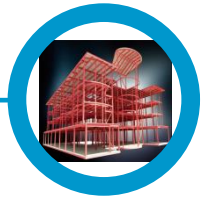
- From the Specifications, we'll find the following requirements:

$$R_n = 0.60 F_y A_{gv}$$ Eq J4-3: Pg-16.1-129

$$R_n = 0.60 F_u A_{nv}$$ Eq J4-4: Pg-16.1-129

  - Note that A_{gv} and A_{nv} are computed in the same fashion as when we discussed block shear (also, in general, $A_{nv} = A_{gv}$).
  - Also:
    - For shear yielding: $\phi = 1.00$
    - For shear fracture: $\phi = 0.75$

Welded Connections



- **Fillet Weld Capacity**

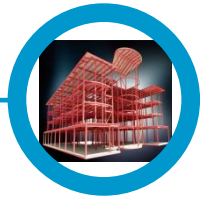
- Analogous to bolt spacing requirements, there are also limits on weld size:
  - Minimum weld sizes:

TABLE J2.4
Minimum Size of Fillet Welds

Material Thickness of Thinner Part Joined, in. (mm)	Minimum Size of Fillet Weld,[a] in. (mm)
To 1/4 (6) inclusive	1/8 (3)
Over 1/4 (6) to 1/2 (13)	3/16 (5)
Over 1/2 (13) to 3/4 (19)	1/4 (6)
Over 3/4 (19)	5/16 (8)

[a] Leg dimension of fillet welds. Single pass welds must be used.
Note: See Section J2.2b for maximum size of fillet welds.

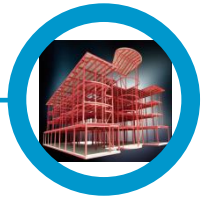
Welded Connections



- **Fillet Weld Capacity**

- The minimum weld size limit is not due to strength but to the quench effect of thick material on small welds:
  - If the weld is too small, the plate will absorb the heat too rapidly
  - This results in rapid cooling of the weld and may result in a loss of weld ductility.

Welded Connections



- **Fillet Weld Capacity**

  - For the maximum weld size limit:

J2b: Pg-16.1-111

$$a_{max} = \begin{cases} t & t < 1/4 \text{ in} \\ t - 1/16 \text{ in} & t \geq 1/4 \text{ in} \end{cases}$$

2b. Limitations

Fillet welds shall meet the following limitations:

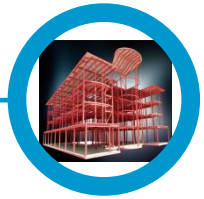
(a) The minimum size of fillet welds shall be not less than the size required to transmit calculated forces, nor the size as shown in Table J2.4. These provisions do not apply to fillet weld reinforcements of PJP or CJP groove welds.

(b) The maximum size of fillet welds of connected parts shall be:

(1) Along edges of material less than 1/4 in. (6 mm) thick; not greater than the thickness of the material.

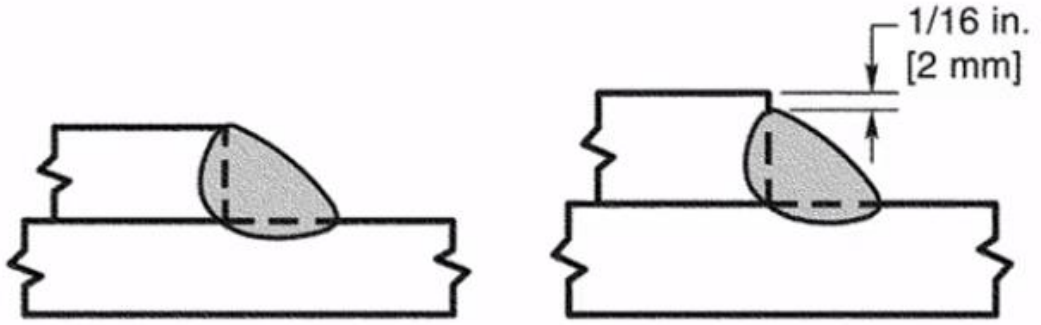
(2) Along edges of material 1/4 in. (6 mm) or more in thickness; not greater than the thickness of the material minus 1/16 in. (2 mm), unless the weld is especially designated on the drawings to be built out to obtain full-throat thickness. In the as-welded condition, the distance between the edge of the base metal and the toe of the weld is permitted to be less than 1/16 in. (2 mm), provided the weld size is clearly verifiable.

Welded Connections



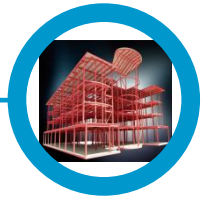
- **Fillet Weld Capacity**

- The reason that we have maximum weld sizes is to try and minimize distortion of plate elements during welding:



MAXIMUM DETAILED SIZE OF FILLET WELD ALONG EDGES

Welded Connections

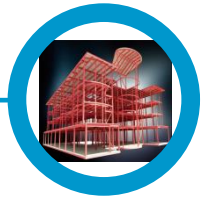


- **Fillet Weld Capacity**

- There are also limits on weld length:
  - The minimum length of a weld is 4a
  - If welds get “too long”, a reduction factor is applied
    - “Too long” begins at 100 times the weld size, “a”

- However, these limits are usually not a major concern.
  - For most building connections, we don’t approach this limit.

Welded Connections



- **Fillet Weld Capacity**

Summary of equations:

– Weld metal capacity:

$$R_n = (0.60F_{EXX})(0.707L_w a)$$
$$\phi = 0.75$$

– Base metal capacity (shear yielding):

$$R_n = 0.60F_y A_{gv}$$
$$\phi = 1.00$$

Base metal capacity (shear fracture):

$$R_n = 0.60F_u A_{nv}$$
$$\phi = 0.75$$

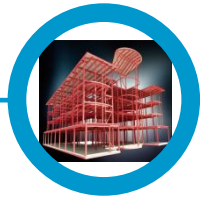
- Summary of weld limits:
  – Minimum weld sizes:

J2b: Pg-16.1-111

  – Maximum weld sizes:

$$a_{\max} = \begin{cases} t & t < \frac{1}{4}\text{ in} \\ t - \frac{1}{16}\text{ in} & t \geq \frac{1}{4}\text{ in} \end{cases}$$

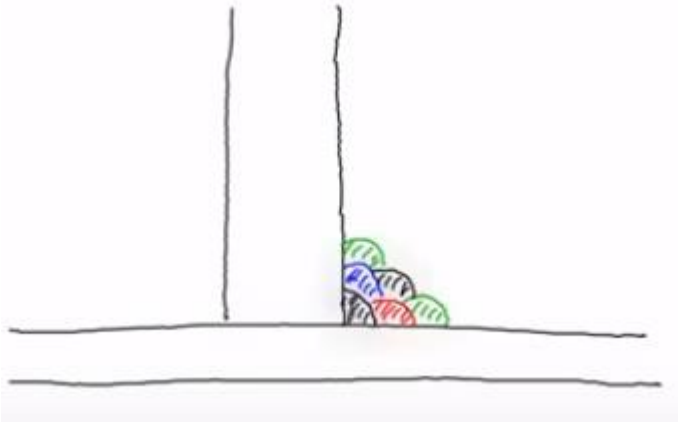
Welded Connections



- **Fillet Weld Capacity**

  - A common weld size used for design is 5/16”
    - The reason why is because this is the largest size weld that can be deposited with a single pass of the electrode.

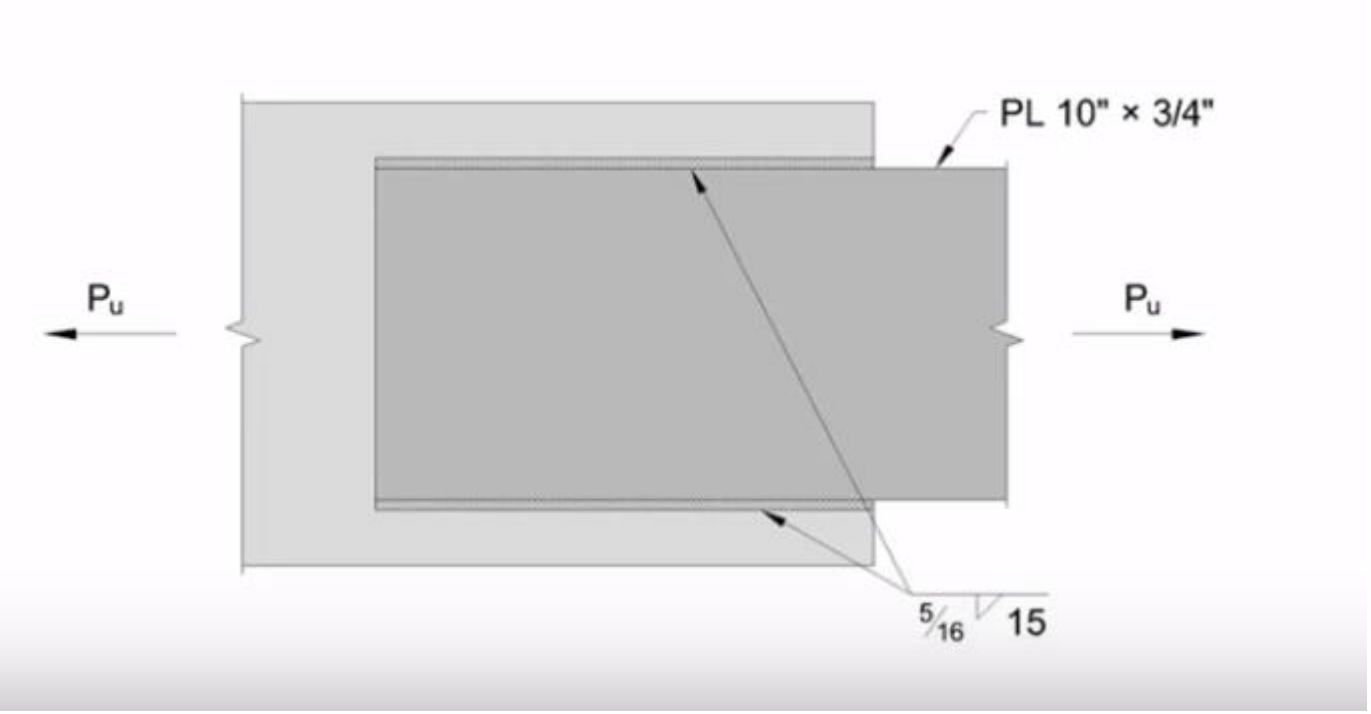
  - When we design, we typically want to choose the largest weld size possible:
    - Results in the shortest weld length
    - Quicker to deposit (i.e. most economical)



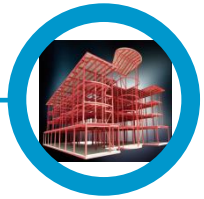
Welded Connections

- **Fillet Weld Capacity**

- Determine the design strength of the connection below.
  - A572 Gr. 50 steel is used for the tension member (PL 10” × ¾”).
  - E70 electrodes are used for the fillet welds.



Welded Connections



- **Fillet Weld Capacity**

- weld metal capacity: $F_{EXX} = 70$ ksi, $a = 5/16"$, $L_w = 2(15) = 30"$

$$\therefore \phi R_n = (0.75)(0.60 F_{EXX})(0.707 a L_w)$$

$$= (0.75)(0.60)(70\text{ ksi})(0.707)(5/16")(30")$$

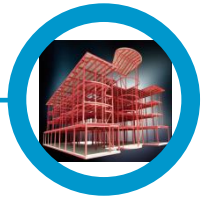
$$\phi R_n = 208.8 \text{ kip}$$

- base metal capacity: $L_w = 30"$, $t = 3/4"$ $\therefore A_{gv} = A_{nv} = L_w t$

$$= (30")(3/4") = 22.5 \text{ in}^2$$

$F_y = 50$ ksi
$F_u = 65$ ksi
[A572, Gr. 50, Tab. 2-4]

Welded Connections



- **Fillet Weld Capacity**

$$\therefore \quad \phi R_{ny} = (1.00)(0.60 F_y A_{gv}) = (1.00)(0.60)(50\,ksi)(22.5\,in^2) = 675.0\,kip$$

$$\phi R_{nf} = (0.75)(0.60 F_u A_{nv}) = (0.75)(0.60)(65\,ksi)(22.5\,in^2) = 658.1\,kip$$

$\therefore$ for the connection:

ANSWER

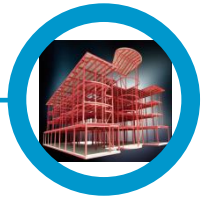
$$\phi R_n = 208.8\,k$$

-weld limits: $a_{min} = 1/4"$ [Tab. J2.4]

$a = 5/16"$ $\therefore$ OK

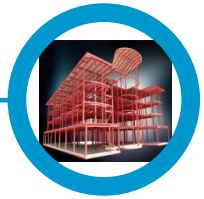
$a_{max} = t - 1/16" = 3/4" - 1/16" = 11/16"$

Welded Connections



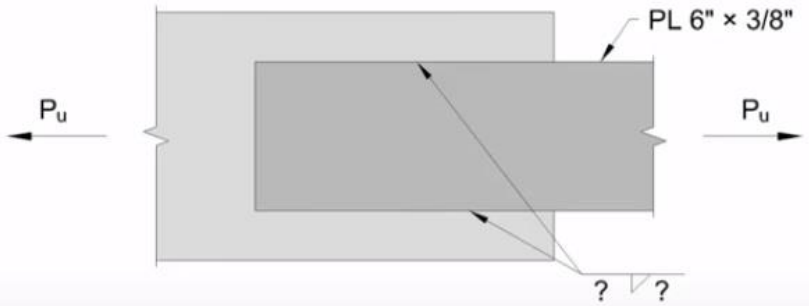
- **Design of Fillet Weld**

- Design of fillet-welded connections is fairly straightforward:
  - Determine the design load.
  - Use weld limitations to determine the appropriate weld size.
  - Determine the design capacity of a one-inch-long weld.
  - Divide the design load by the capacity to obtain the length of weld required.

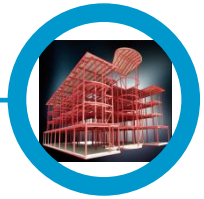


- **Design of Fillet Weld**

- Using E70 electrodes, design a fillet weld to resist a service dead load of 15 kips and a service live load of 40 kips.
  - A572 Gr. 50 steel plate is used.
  - Assume the capacity of the gusset plate to be adequate.



Welded Connections



- **Design of Fillet Weld**

- factored load: $P_u = 1.2P_D + 1.6P_L = 1.2(15\,k) + 1.6(40^k) = 82^k$

- weld limits: $\{t = 3/8"\}$

$a_{min} = 3/16"$ (Tab. J2.4, Pg. 16.1-111)

$a_{max} = t - 1/16" = 3/8" - 1/16" = 5/16"$

Use $a = 5/16"$

- compute the capacity of 1" of weld: $a = 5/16"$, $F_{EXX} = 70\,ksi$, $L_w = 1"$, $t = 3/8"$

$F_y = 50\,ksi$, $F_u = 65\,ksi$ (Tab. 2-4)

$A_{gv} = A_{nv} = L_w t = 0.375\,in^2$

- weld metal capacity: $\phi r_n = \phi(0.60F_{EXX})(0.707aL_w)$

$= (0.75)(0.60)(70\,ksi)(0.707)(5/16")(1") = 6.96\,kips$ for 1" of weld

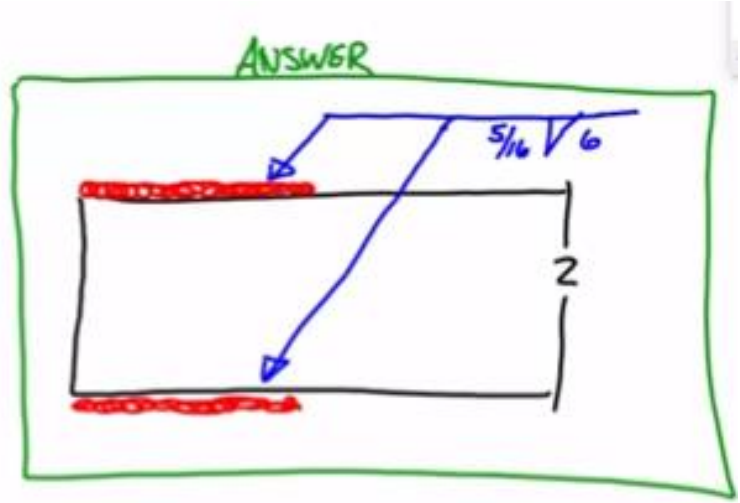
- base metal yielding: $\phi r_n = \phi(0.60F_y A_{gv}) = (1.00)(0.60)(50\,ksi)(0.375\,in^2) = 11.25\,kips$ for 1" of weld

- base metal fracture: $\phi r_n = \phi(0.60F_u A_{nv}) = (0.75)(0.60)(65\,ksi)(0.375\,in^2) = 10.97\,kips$ for 1" of weld

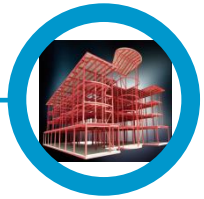
Welded Connections

- **Design of Fillet Weld**

$$\therefore \; L_{w_{req'd}} = \frac{P_u}{\phi r_n} = \frac{82 \text{ kip}}{6.96 \text{ kip/inch}} = 11.78''$$



Welded Connections

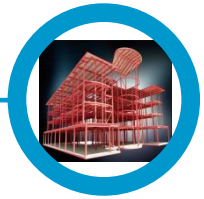


- **Design of Fillet Weld**

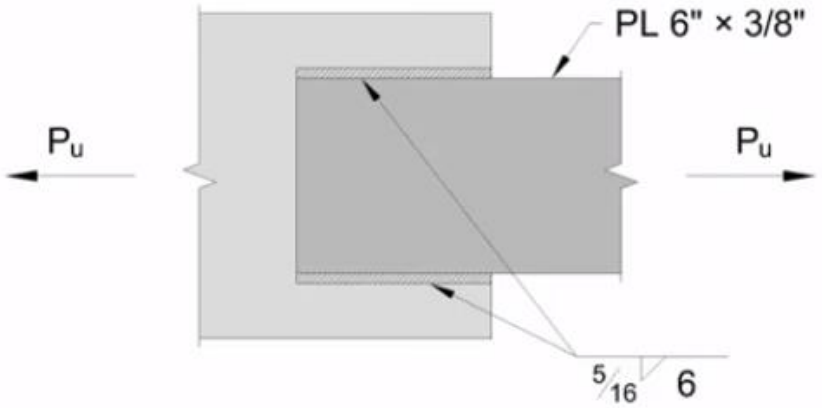
  - This should be intuitive for a member like a plate.
  - However, what about an angle?
    - In an angle, the centroid isn't located at half the distance from the edge.

  - We would then want to place welds such that any bending wouldn't occur.
    - This is known as a balanced weld arrangement.

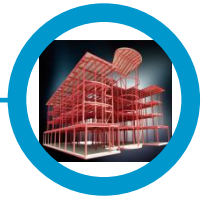
Welded Connections



- **Design of Fillet Weld**
  - Let's look back at the weld design of our previous example:
    - Two longitudinal welds of even length.

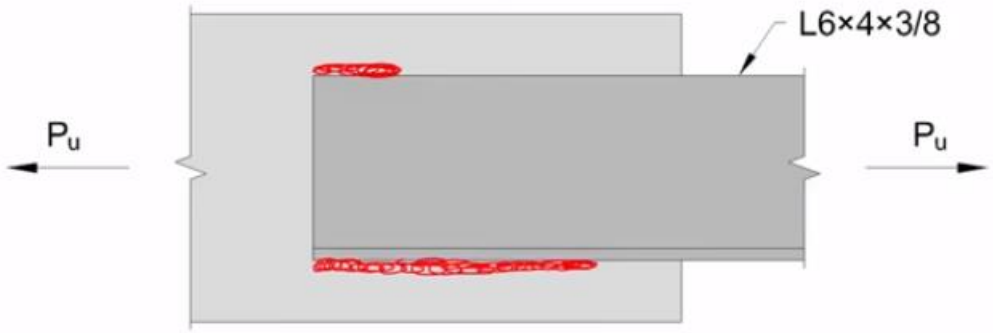


    - This weld configuration seemed to make sense:
      - Resisted the axial load from the plate
      - Eliminated any bending from load eccentricities

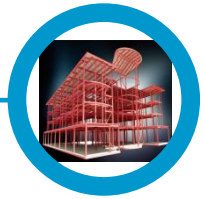


- **Design of Fillet Weld**

- Design balanced fillet welds to resist a design load of P_u = 150 kip.
  - Assume the capacity of the gusset plate to be adequate.
  - Also, assume that A572 Gr. 50 steel and E70 electrodes are used.



Welded Connections



- **Design of Fillet Weld**

- factored load: $P_u = 150^k$

- material properties: $F_{EXX} = 70$ ksi, $F_y = 50$ ksi, $F_u = 65$ ksi

- weld limits: $a_{min} = 3/16"$

$a_{max} = t - 1/16" = 3/8" - 1/16" = 5/16"$

Use $a = 5/16"$

$\therefore$ weld metal capacity $= 6.96^{kip/in}$

base metal yielding $= 11.25^{kip/in}$

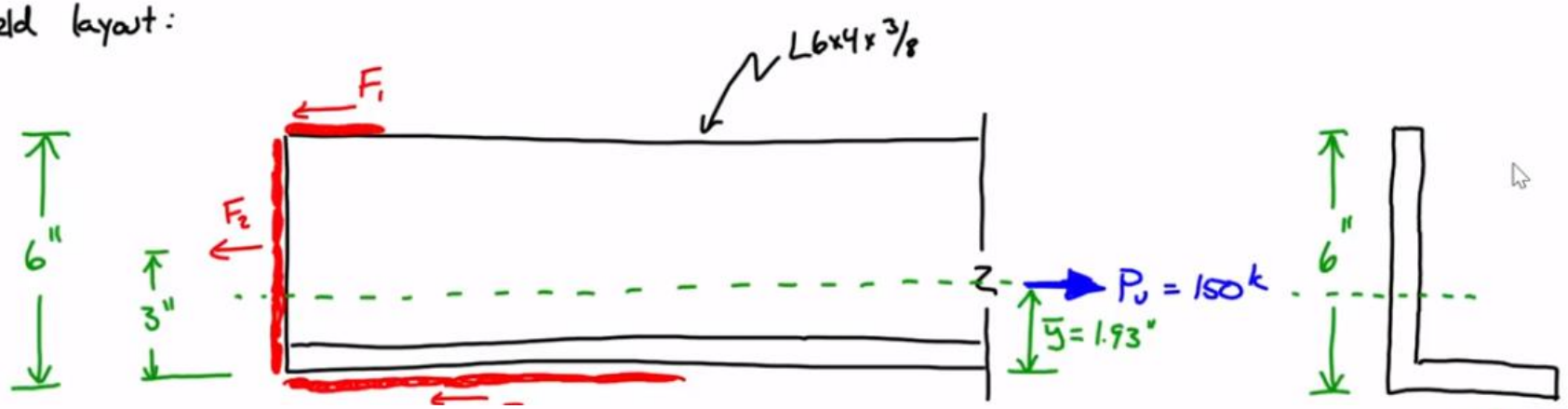
base metal fracture $= 10.97^{kip/in}$

$\phi r_n = 6.96^{kip/in}$

Welded Connections

- **Design of Fillet Weld**

- weld layout:



$F_2 = (6.96\ ^{kip}/_{in})(6") = 41.76^k$

$\Sigma M = 0$

↺	↻
$P_u(\bar{y})$	$F_1(6")$
	$F_2(3")$

$P_u \bar{y} = F_1(6") + F_2(3")$

$$\therefore F_1 = \frac{P_u \bar{y} - F_2(3")}{6"} = \frac{(150^k)(1.93") - (41.76^k)(3")}{6"} = 27.37^k$$

$$L_1 = \frac{F_1}{\phi r_n} = \frac{27.37^k}{6.96\ ^k/_{in}} = 3.93"$$

Use $L_1 = 4"$

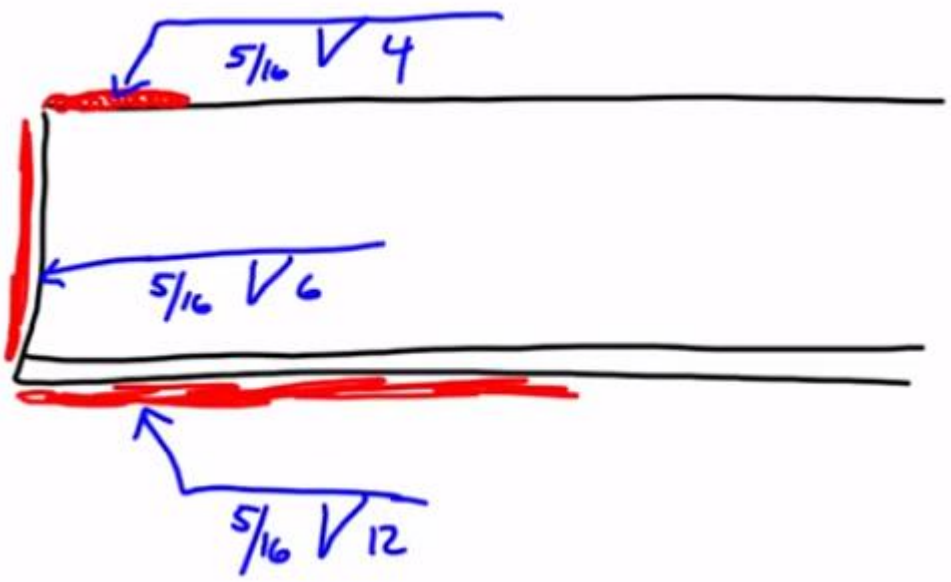
Welded Connections

- **Design of Fillet Weld**

$\Sigma F = 0$: $F_3 = P_v - F_1 - F_2 = 150k - 41.76^k - 27.37^k = 80.87^k$

$L_3 = F_3 / \phi r_n = 11.60"$

Use $L_3 = 12"$




Questions?